uniformly before you flailed your arms, and they will be spread uniformly after you're done. The uniformly spread gas configuration is insensitive to an enormous number of rearrangements of its molecular constituents, and so is in a state of high entropy. By contrast, if the gas were spread in a smaller space, as when it was in the bottle, or confined by a barrier to a corner of the room, it has significantly lower entropy. The reason is simple. Just as thinner books have fewer page reorderings, smaller spaces provide fewer places for molecules to be located, and so allow for fewer rearrangements.

But when you twist off the bottle's cap or remove the barrier, you open up a whole new universe to the gas molecules, and through their bumping and jostling they quickly disperse to explore it. Why? It's the same statistical reasoning as with the pages of *War and Peace*. No doubt, some of the jostling will move a few gas molecules purely within the initial blob of gas or nudge a few that have left the blob back toward the initial dense gas cloud. But since the volume of the room exceeds that of the initial cloud of gas, there are *many* more rearrangements available to the molecules if they disperse out of the cloud than there are if they remain within it. On average, then, the gas molecules will diffuse from the initial cloud and slowly approach the state of being spread uniformly throughout the room. Thus, the lower-entropy initial configuration, with the gas all bunched in a small region, naturally evolves toward the higher-entropy configuration, with the gas uniformly spread in the larger space. And once it has reached such uniformity, the gas will tend to maintain this state of high entropy: bumping and jostling still causes the molecules to move this way and that, giving rise to one rearrangement after another, but the overwhelming majority of these rearrangements do not affect the gross, overall appearance of the gas. That's what it means to have high entropy.[9]

In principle, as with the pages of *War and Peace*, we could use the laws of classical physics to determine precisely where each carbon dioxide molecule will be at a given moment of time. But because of the enormous number of CO_2 molecules—about 10^{24} in a bottle of Coke—actually carrying out such calculations is practically impossible. And even if, somehow, we were able to do so, having a list of a million billion billion particle positions and velocities would hardly give us a sense of